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DEEMED-TO-BE UNIVERSITY

SCHOOL OF
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SCIENCE AND IT

Andriod Operating System

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UNIT3:Introduction to Android

Android Architecture:

Android architecture contains different number of components to support any android device needs. Android software contains an open-source Linux Kernel having collection of number of C/C++ libraries which are exposed through an application framework services.

Among all the components Linux Kernel provides main functionality of operating system functions to smartphones and Dalvik Virtual Machine (DVM) provide platform for running an android application.

The main components of android architecture are following:-

Applications

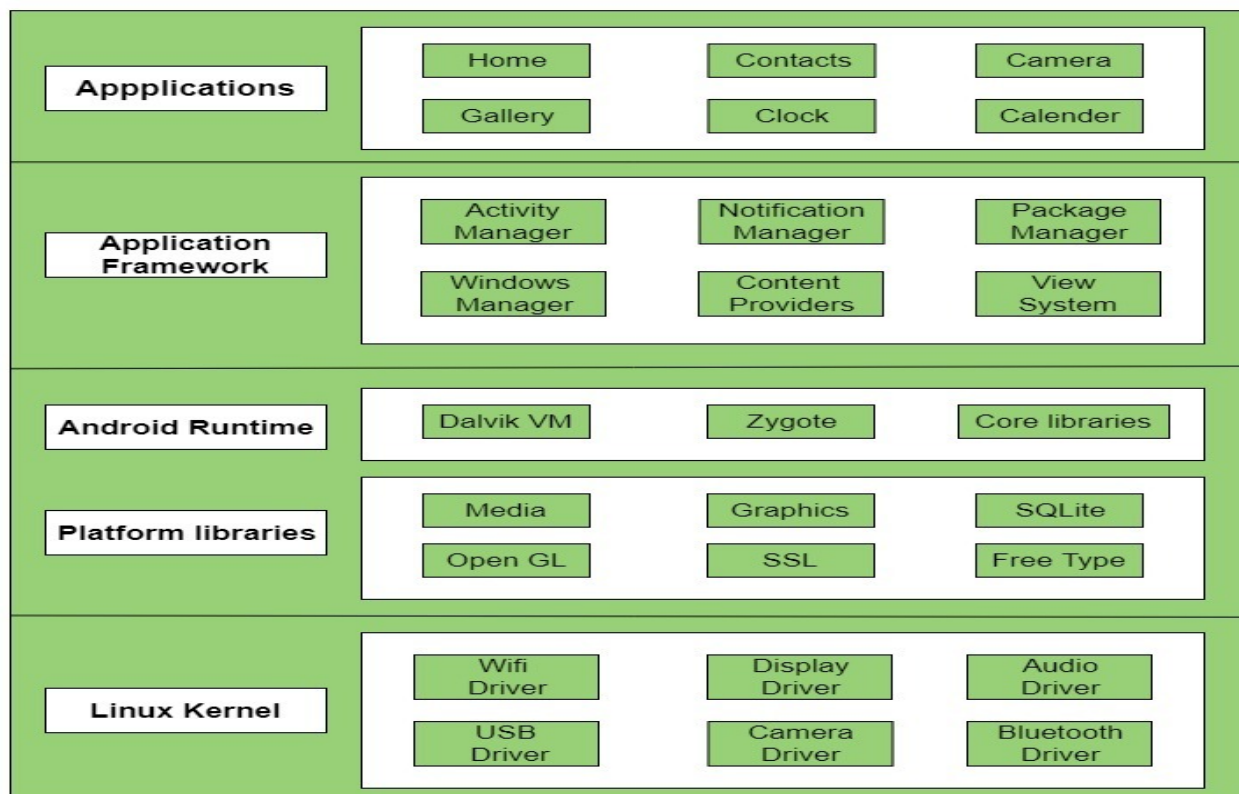
Application Framework

Android Runtime

Platform Libraries

Linux Kernel(kernel is a part of the operating system that converts user commands into machine language)

- Pictorial representation of android architecture with several main components and their sub components –



Applications –

Applications is the top layer of android architecture. The pre-installed applications like home, contacts, camera, gallery etc and third party applications downloaded from the play store like chat applications, games etc. will be installed on this layer only. It runs within the Android run time with the help of the classes and services provided by the application framework.

Application framework –

Application Framework provides several important classes which are used to create an Android application. It provides a generic abstraction for hardware access and also helps in managing the user interface with application resources. Generally, it provides the services with the help of which we can create a particular class and make that class helpful for the Applications creation.

It includes different types of services activity manager, notification manager, view system, package manager etc. which are helpful for the development of our application according to the prerequisite.

Application runtime –

Android Runtime environment is one of the most important part of Android. It contains components like core libraries and the Dalvik virtual machine(DVM). Mainly, it provides the base for the application framework and powers our application with the help of the core libraries.

Like Java Virtual Machine (JVM), **Dalvik Virtual Machine (DVM)** is a register-based virtual machine and specially designed and optimized for android to ensure that a device can run multiple instances efficiently. It depends on the layer Linux kernel for threading and low-level memory management. The core libraries enable us to implement android applications using the standard JAVA or Kotlin programming languages.

Platform libraries –

The Platform Libraries includes various C/C++ core libraries and Java based libraries such as Media, Graphics, Surface Manager, OpenGL etc. to provide a support for android development.

Media library provides support to play and record an audio and video formats.

Surface manager responsible for managing access to the display subsystem.

(Simple Graphics Library)SGL and **OpenGL** both cross-language, cross-platform application program interface (API) are used for 2D and 3D computer graphics.

SQLite provides database support and **FreeType** provides font support.

Web-Kit This open source web browser engine provides all the functionality to display web content and to simplify page loading.

SSL (Secure Sockets Layer) is security technology to establish an encrypted link between a web server and a web browser.

Linux Kernel –

Linux Kernel is heart of the android architecture. It manages all the available drivers such as display drivers, camera drivers, Bluetooth drivers, audio drivers, memory drivers, etc. which are required during the runtime.

The Linux Kernel will provide an abstraction layer between the device hardware and the other components of android architecture. It is responsible for management of memory, power, devices etc.

The features of Linux kernel are:

Security: The Linux kernel handles the security between the application and the system.

Memory Management: It efficiently handles the memory management thereby providing the freedom to develop our apps.

Process Management: It manages the process well, allocates resources to processes whenever they need them.

Network Stack: It effectively handles the network communication.

Driver Model: It ensures that the application works properly on the device and hardware manufacturers responsible for building their drivers into the Linux build.



History of Android

The history and versions of android are interesting to know. The code names of android ranges from A to J currently, such as **Aestro, Blender, Cupcake, Donut, Eclair, Froyo, Gingerbread, Honeycomb, Ice Cream Sandwich, Jelly Bean, KitKat** and **Lollipop**.

- 1) Initially, **Andy Rubin** founded Android Incorporation in Palo Alto, California, United States in October, 2003.
- 2) In 17th August 2005, Google acquired android Incorporation. Since then, it is in the subsidiary of Google Incorporation.
- 3) The key employees of Android Incorporation are **Andy Rubin, Rich Miner, Chris White** and **Nick Sears**.
- 4) Originally intended for camera but shifted to smart phones later because of low market for camera only.
- 5) Android is the nick name of Andy Rubin given by coworkers because of his love to robots.
- 6) In 2007, Google announces the development of android OS.
- 7) In 2008, HTC launched the first android mobile.

Android Versions, Codename and API

Let's see the android versions, codenames and API Level provided by Google.

Version	Code name	API Level
1.5	Cupcake	3
1.6	Donut	4
2.1	Eclair	7
2.2	Froyo	8
2.3	Gingerbread	9 and 10
3.1 and 3.3	Honeycomb	12 and 13
4.0	Ice Cream Sandwich	15
4.1, 4.2 and 4.3	Jelly Bean	16, 17 and 18
4.4	KitKat	19
5.0	Lollipop	21
6.0	Marshmallow	23
7.0	Nougat	24-25
8.0	Oreo	26-27

Key Features of the Android Operating System

User Interface: The user interface of the Android operating system is straight forward, and these features make it very user friendly.

Multiple Language Support: Android supports multiple languages in its operating system and one can change the language very easily based on one's requirement, the International languages supported are English, Germany, Chinese, Dutch, French, German, Japanese, Korean, Russian, and many more also some native language of India is also Supported Like Hindi, Marathi, Gujrati, Punjabi and many more.

Multi-tasking: Android provides support to run apps and services in the background with ease which allows the users to use multiple apps at the same time.

Connectivity: Android has extensive support to the connectivity and it supports connectivity such as WiFi, Bluetooth, Hotspot, CDMA, GSM, NFC, VOLTE, UBB, VPN, 3G network band, and 4G Network Band.

Extensive Application Support: Android have Play store which is used as the major tool to download and update applications on the operating system, however, one can download the installer(often called as APK file) and install it manually, but it is not much recommended as third party applications could be prone to some security breach in the smartphones.

Key Features of Android

1. **Smart Replies:** Automatic Suggestion to reply to texting app, like if a message contains a phone number then we can click on it and save/call the contact number. In this feature, you not only get some suggested responses but also get recommended actions related to the message. For Example, if someone sends a message regarding a meeting schedule then it pops to set a reminder for the meeting in the calendar, If one get a location in the message then it pops to open maps for it, If someone asks to join for a meeting then it prompts us to reply with yes or no.

2. Sound Amplifier

Native Support for the inbuilt amplifier for the users, earlier we had to download some third-party applications to do the same. With Sound Amplifier feature one can hear the better sound and can also boost the sound by fine-tuning the sound by the help of features such as “boost” and “fine-tune” on either of the ears or both of the ears of the headphone. This feature helps the user to hear the music/recorded voice note more clearly when a large amount of hindrance can be observed in the surrounding.

3. Gesture Control

Earlier this feature was not only available for IOS Applications but with the android 10 update, we can get the use it on the windows operating system. Some popular gesture control features include:

Go back: In order to go back one needs to swipe from the left or right edge of the screen, and while doing so an arrow shows up on the screen showing that you are redirected to the previous page of the device.

Go home: To go to the home screen one needs to swipe from the bottom edge of the screen in the upward direction and you will be redirected to the home screen.

4. Live Captions

Using machine learning algorithms to generate the live captions of any video you are watching even when the video don't have subtitles. To turn Live Caption on or off we need to Press the volume button and in the volume control one needs to tap on the live caption subtitles

5. Dark Mode

The dark mode is provided in the operating system and the new dark mode support many apps that will reduce the battery drain on OLED phones. Android dark theme uses true black to keep the battery alive for a longer duration, also it changes the view of the google applications like Google Drive, Google Photos, Google Calendar, to improve the viewing experience at night and to put less strains on the eyes for night users.

Changes Made for Android Applications Developers

Increased Privacy:

- Changed background execution limit.
- Wifi Scanning programmatically disabled.
- Can not toggle mobile data Programmatically.
- Removed access to IEMI codes.

Limited Storage:

- Does not allow apps to access entire SD card.
- Uses scopes like file type differentiator like pictures, music, etc.

Camera:

- Restriction on access to camera details.
- Can not access camera metadata directly.

Biometric Access:

- Access to check if device has biometric into it.
- Use biometric authentication in android application.

Audio Device Sharing:

- Multiple audio devices could use the audio device at the same time.
- You can call someone and use other audio device also at the same time.

Android SDK and it's Components

Android SDK stands for **Android Software Development Kit** which is developed by Google for Android Platform. With the help of Android SDK, we can create android Apps easily.

About Android SDK

Android SDK is a collection of libraries and Software Development tools that are essential for Developing Android Applications. Whenever Google releases a new version or update of Android Software, a corresponding SDK also releases with it. In the updated or new version of SDK, some more features are included which are not present in the previous version. Android SDK consists of some tools which are very essential for the development of Android Application. These tools provide a smooth flow of the development process from developing and debugging. Android SDK is compatible with all operating systems such as Windows, Linux, macOS, etc.

1. Android SDK Tools

Android SDK tool is an important component of Android SDK. It consists of a complete set of development and debugging tools. Below are the SDK developer tools:

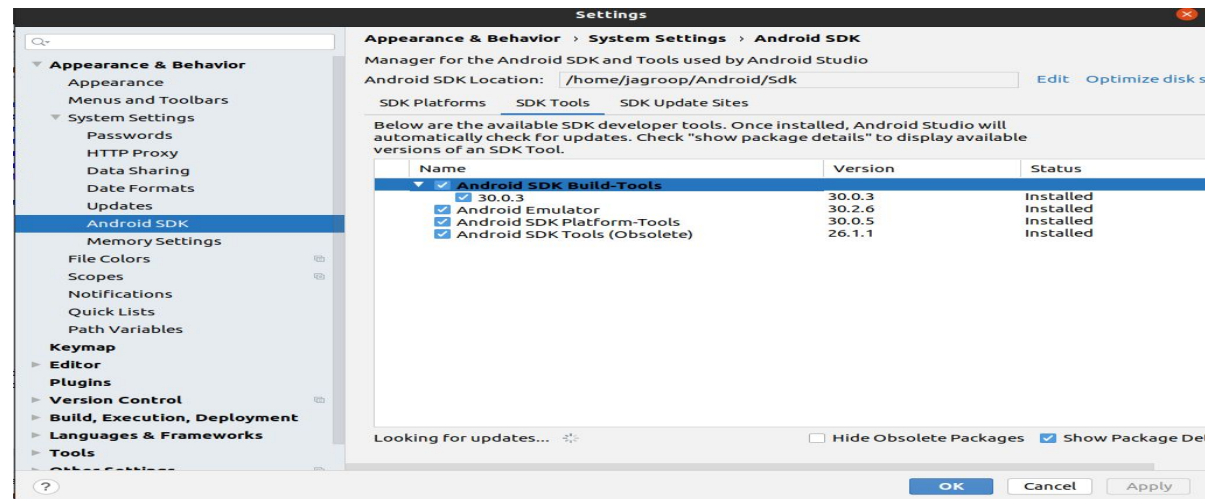
Android SDK Build tool.

Android Emulator.

Android SDK Platform-tools.

Android SDK Tools.

These are shown below :



2. Android SDK Build-Tools

Android SDK build tools are used for building actual binaries of Android App. The main functions of Android SDK Build tools are built, debug, run and test Android applications. The latest version of the Android SDK Build tool is "34.0.0". While downloading or updating Android in our System, one must ensure that its latest version is download in SDK Components.

3. Android Emulator

An Android Emulator is a device that simulates an Android device on your system. Suppose we want to run our android application that we code. One option is that we will run this on our Android Mobile by Enabling USB Debugging on our mobile. Another option is using Android Emulator. In Android Emulator the virtual android device is shown on our system on which we run the Android application that we code.

In Android Virtual Emulator all functions that are feasible on real Android mobile is works on virtual Device like:

phone calls, text messages.

stimulate different network speeds.

specify the location of a device

access on google play store and lot's more.

But there is one disadvantage of this emulator is that. It is very slow when System's PC has less RAM. It works fine when a maximum GB of RAM is present on our device.

4. Android SDK Platform-tools

Android SDK Platform-tools is helpful when we are working on Project and they will show the error messages at the same time. It is specifically used for testing. It includes:

Android Debug Bridge (ADB), is a command-line tool that helps to communicate with the device. It allows us to perform an action such as Installing App and Debugging App etc.

Fastboot allows you to flash a device with a new system image.

Systrace tools help to collect and inspect timing information. It is very crucial for App Debugging.

5. Android SDK Tools

Android SDK tool is a component of SDK tool. It consists of a set of tools which and other Utilities which are crucial for the development of Android Application. It contains the complete set of Debugging and Development tools for android.

6. SDK Platforms

For Each Android Software, one SDK platform is available as shown below:

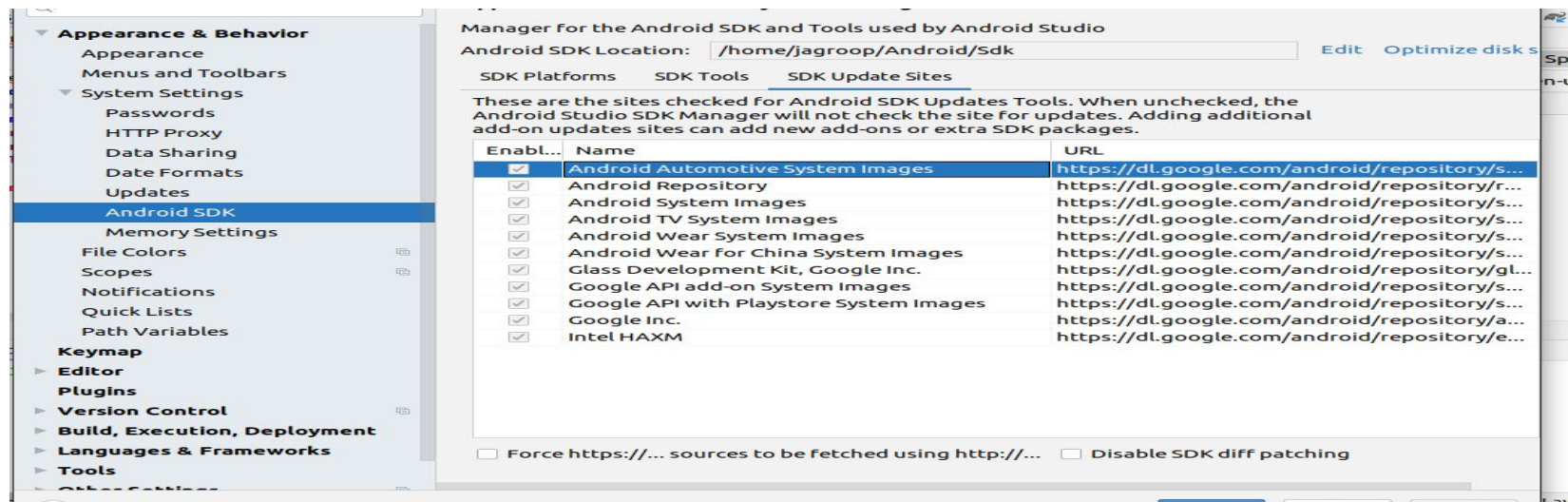
Like Android 14.0.

These are numbered according to the android version. The new version of the SDK platform has more features and more compatible but the old version is less compatible with fewer features. Like in Android 14.0 have more compatible and have more feature but the below versions like Android 13.0.

7. SDK Update Sites

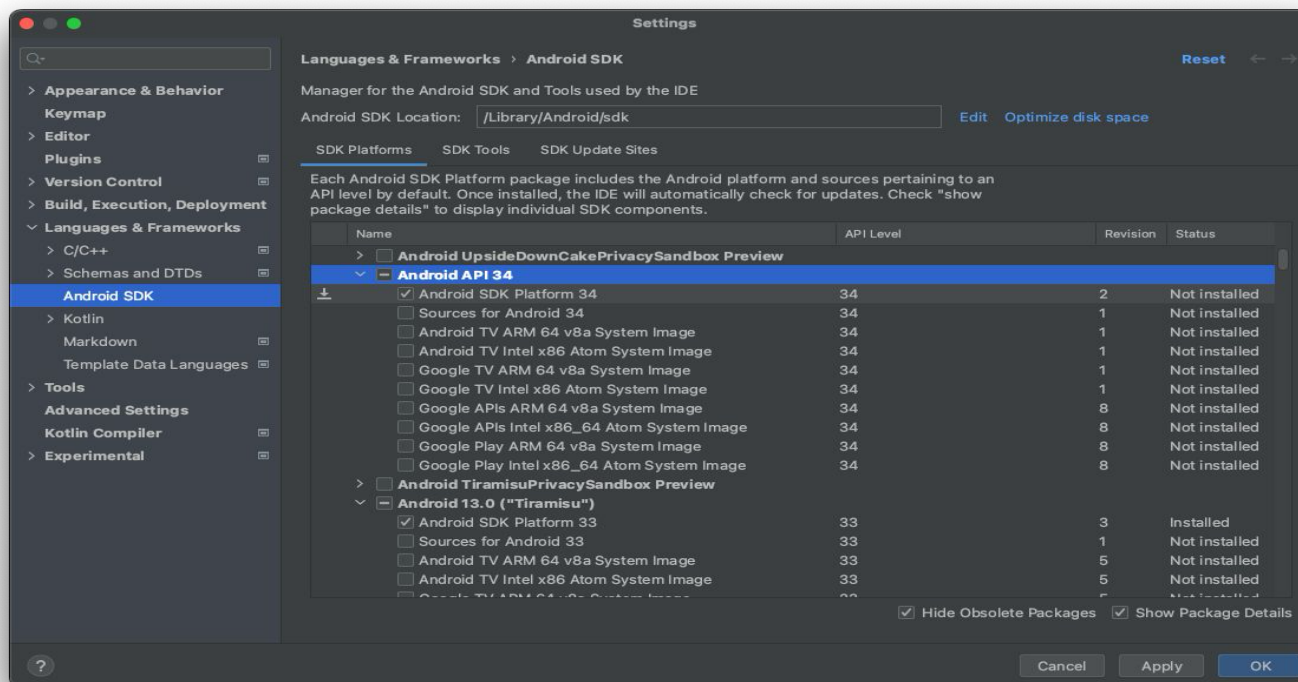
In SDK Update Sites, some sites are embedded in it which will check for Android SDK

In SDK Update Sites, some sites are embedded in it which will check for Android SDK Updates Tools. In this, one must ensure we don't unclick the button below because these are checked by default which will check for updates if we will unclick it then it doesn't check updates for those.



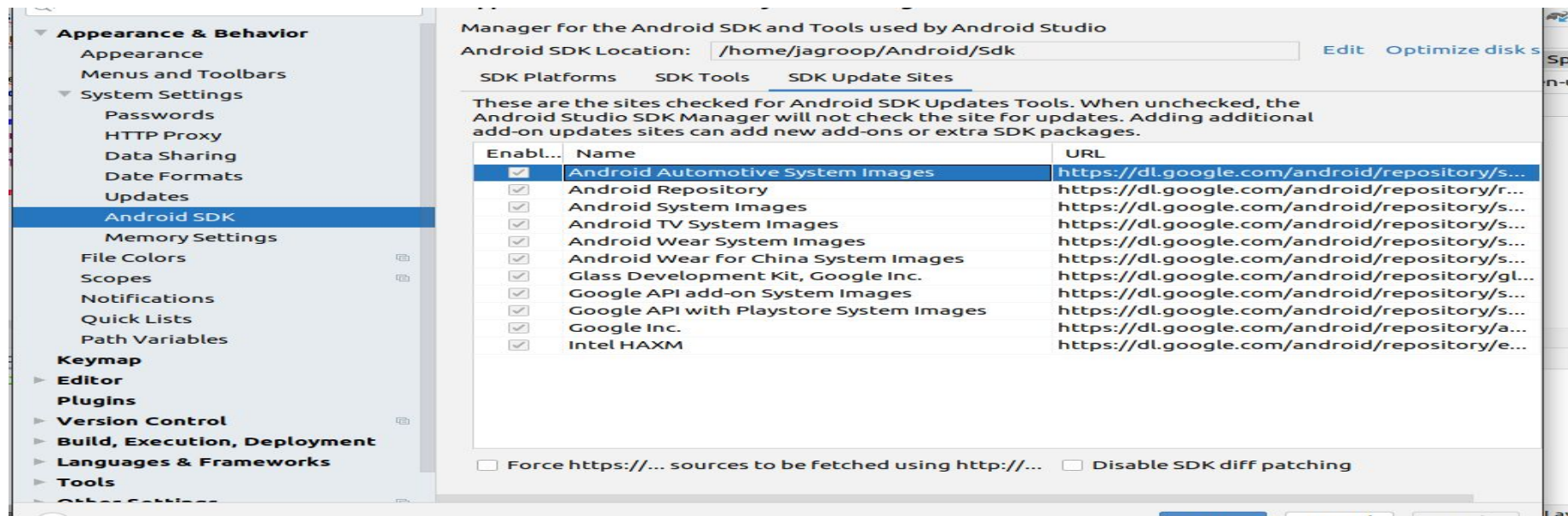
8. SDK Platforms

- For Each Android Software, one SDK platform is available as shown below:



- 7. SDK Update Sites

In SDK Update Sites, some sites are embedded in it which will check for Android SDK Updates Tools. In this, one must ensure we don't unclick the button below because these are checked by default which will check for updates if we will unclick it then it doesn't check updates for those.



- **The Application Manifest File**

- Every project in Android includes a manifest file, which is **AndroidManifest.xml**, stored in the root directory of its project hierarchy. The manifest file is an important part of our app because it defines the structure and metadata of our application, its components, and its requirements. This file includes nodes for each of the [Activities](#), [Services](#), [Content Providers](#), and [Broadcast Receiver](#) that make the application and using [Intent Filters](#) and Permissions determines how they co-ordinate with each other and other applications. A manifest file includes the nodes that define the application components, security settings, test classes, and requirements that make up the application. Some of the manifest sub-node tags that are mainly used are:

1. uses-sdk

It is used to define a minimum and maximum SDK version that must be available on a device so that our application functions properly, and target SDK for which it has been designed using a combination of minSdkVersion, maxSdkVersion, and targetSdkVersion attributes, respectively.

2. uses-configuration

The uses-configuration nodes are used to specify the combination of input mechanisms that are supported by our application. It is useful for games that require particular input controls.

3. uses-features

It is used to specify which hardware features your application requirement. This will prevent our application from being installed on a device that does not include a required piece of hardware such as NFC hardware, as follows:

```
<uses-feature android:name="android.hardware.nfc"/>
```

4. supports-screens

It is used to describe the screen support for our application:

```
<supports-screens  
  android:smallScreens="false"  
  android:normalScreens="true"  
  android:largeScreens="true"  
  android:xlargeScreens="true"/>
```

5. permission

It is used to create permissions to restrict access to shared application components. We can also use the existing platform permissions for this purpose or define your own permissions in the manifest.

- **6. application**

- A manifest can contain only one application node. It uses attributes to specify the metadata for your application (including its title, icon, and theme). During development, we should include a debuggable attribute set to true to enable debugging, then be sure to disable it for your release builds. The application node also acts as a container for the Activity, Service, Content Provider, and Broadcast Receiver nodes that specify the application components.
- <application
- android:icon="@drawable/icon"
- android:logo="@drawable/logo"
- android:theme="@android:style/Theme.Light"
- android:name=".MyApplicationClass"
- android:debuggable="true">
- [...application nodes.....]
- </application>

Android - Application Components

Sr.No	Components & Description
1	Activities They dictate the UI and handle the user interaction to the smart phone screen.
2	Services They handle background processing associated with an application.
3	Broadcast Receivers They handle communication between Android OS and applications.
4	Content Providers They handle data and database management issues.

Activity

An activity represents a single screen with a user interface, in short Activity performs actions on the screen. For example, an email application might have one activity that shows a list of new emails, another activity to compose an email, and another activity for reading emails. If an application has more than one activity, then one of them should be marked as the activity that is presented when the application is launched.

An activity is implemented as a subclass of **Activity** class as follows –

```
public class MainActivity extends Activity { }
```

Manifest File

The **AndroidManifest.xml file** *contains information of your package* , including components of the application such as activities, services, broadcast receivers, content providers etc.

It performs some other tasks also:

It is **responsible to protect the application** to access any protected parts by providing the permissions.

It also **declares the android api** that the application is going to use.

It **lists the instrumentation classes** . The instrumentation classes provides profiling and other informations. These informations are removed just before the application is published etc.

This is the required xml file for all the android application and located inside the root directory.



Programming paradigms and Application Components Part 2: Intents

Android Intent is the *message* that is passed between components such as activities, content providers, broadcast receivers, services etc. It is generally used with startActivity() method to invoke activity, broadcast receivers etc.

The **dictionary meaning** of intent is *intention or purpose*. So, it can be described as the intention to do action.

Android intents are mainly used to:

- Start the service
- Launch an activity
- Display a web page
- Display a list of contacts
- Broadcast a message
- Dial a phone call etc.

Content providers

Content Provider facilitates access to a central data store or warehouse to allow data sharing and data manipulation across different applications.

Here are a few examples of default Content Providers in Android system's API: These content providers allow the user abstraction from an underlying database. Generally, Content Providers use SQLite to store underlying database.

Call Logs	Provides access to Missed Calls, Call Details (time of call, duration of call)
Address Book	Provides Contact Details
User Dictionary	Keeps track of the words added to the default dictionary



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Broadcast receivers

Broadcast in android is the system-wide events that can occur when the device starts, when a message is received on the device or when incoming calls are received, or when a device goes to airplane mode, etc. Broadcast Receivers are used to respond to these system-wide events. Broadcast Receivers allow us to register for the system and application events, and when that event happens, then the register receivers get notified. There are mainly two types of Broadcast Receivers:

Static Broadcast Receivers: These types of Receivers are declared in the manifest file and works even if the app is closed.

Dynamic Broadcast Receivers: These types of receivers work only if the app is active or minimized.



Services

Android service is a component that is *used to perform operations on the background* such as playing music, handle network transactions, interacting content providers etc. It doesn't has any UI (user interface). The service runs in the background indefinitely even if application is destroyed.

Moreover, service can be bounded by a component to perform interactivity and inter process communication (IPC).